



NORTH SOUTH UNIVERSITY

Committed to the highest standards of academic excellence

School of Engineering and Physical Science Department of Mathematics & Physics

Assignment-01

Semester: **Spring 2023**

Course Coordinator: **Dr. Mohammad Monir Uddin**

Email: monir.uddin@northsouth.edu

Course Name and Course Code:

Introduction to Linear Algebra, MAT-125

Student's Name: **Israt Zaman Srity (2211084042)**

Date: **21/02/2023**

The implication of matrix in real life

Here I tried to show a simplified version of the implication of a matrix in our real life; the information is given below:-

=> here's some information about a garment company's three factories, like how many shirts they produce in a month, how much money they spend on the production, and how much profit.

We have information about three garment factories (M, N, P) and three types of shirts(H, M, L) they produce, but here, some variation occurs in production and raw materials. So the amount of production and the price of shirts also differ based on quality.

FACTORIES NAME / AMOUNT OF SHIRTS	H	M	L
M	2000	1500	1000
N	1800	1700	1500
P	3000	2500	2000

And the production costs of the three factories are 120, 115 and 110.

The selling price of the three factories are the same, and they are 1500, 1200 and 1000.

ANS:-

Let,

$$\begin{bmatrix} 2000 & 1500 & 1000 \\ 1800 & 1700 & 1500 \\ 3000 & 2500 & 2000 \end{bmatrix}$$

The number of shirts, **A**=

$$\text{Production cost } \mathbf{P} = \begin{bmatrix} 120 \\ 115 \\ 110 \end{bmatrix}$$

$$\text{Selling price, } \mathbf{S} = \begin{bmatrix} 1500 \\ 1200 \\ 1000 \end{bmatrix}$$

Now, to find the total production cost,

$$\begin{aligned} \mathbf{A} \times \mathbf{P} &= \begin{bmatrix} 2000 & 1500 & 1000 \\ 1800 & 1700 & 1500 \\ 3000 & 2500 & 2000 \end{bmatrix} \times \begin{bmatrix} 120 \\ 115 \\ 110 \end{bmatrix} \\ &= \begin{bmatrix} 2000 \cdot 120 + 1500 \cdot 115 + 1000 \cdot 110 \\ 1800 \cdot 120 + 1700 \cdot 115 + 1500 \cdot 110 \\ 3000 \cdot 120 + 2500 \cdot 115 + 2000 \cdot 110 \end{bmatrix} \\ &= \begin{bmatrix} 522500 \\ 576500 \\ 867500 \end{bmatrix} \end{aligned}$$

So, the total cost is =(522500+576500+867500)

$$=1,966,500 \text{ tk}$$

And for the profit :

Profit, I = selling price - production cost

$$= \begin{bmatrix} 1500 \\ 1200 \\ 1000 \end{bmatrix} - \begin{bmatrix} 120 \\ 115 \\ 110 \end{bmatrix}$$

$$= \begin{bmatrix} 1380 \\ 1085 \\ 890 \end{bmatrix}$$

$$\text{Now, } A \times I = \begin{bmatrix} 2000 & 1500 & 1000 \\ 1800 & 1700 & 1500 \\ 3000 & 2500 & 2000 \end{bmatrix} \times \begin{bmatrix} 1380 \\ 1085 \\ 890 \end{bmatrix}$$

$$= \begin{bmatrix} 2000 \cdot 1380 + 1500 \cdot 1085 + 1000 \cdot 890 \\ 1800 \cdot 1380 + 1700 \cdot 1085 + 1500 \cdot 890 \\ 3000 \cdot 1380 + 2500 \cdot 1085 + 2000 \cdot 890 \end{bmatrix}$$

$$= \begin{bmatrix} 5277500 \\ 5663500 \\ 8632500 \end{bmatrix}$$

So, the total profit they gain = $(5277500+5663500+8632500)$
 $=19,573,500 \text{ tk}$

(ans)